Shubham Jadhav | CS24S009

Department of Computer Science & Engineering, IIT Madras

Portfolio: shubham-jadhav-FiatLux.github.io
GitHub: github.com/shubham-jadhav-FiatLux
Email: cs24s009@cse.iitm.ac.in

Education

Year	Degree / Certificate	Institute	CGPA / %
2024 – Present	M.S. by Research / CSE	Indian Institute of Technology Madras	7.78 / 10
2017 – 2021	B.Tech / CSE	Savitribai Phule Pune University	9.11 / 10
2017	Class XII (Maharashtra State Board)	Vivekanand College Kolhapur	82.23 %
2015	Class X (Maharashtra State Board)	Gadhinglaj High School Gadhinglaj	94.20 %

Technical Skills

Languages	C and C++, Python, Java, SQL
HPC	CUDA, OpenMP, OpenACC, MPI, HIP
Rendering	OpenGL, WebGL, DirectX, Win32 SDK
Tools	Linux/POSIX, LLVM, Docker, Ghidra, Git/GitHub, LaTeX, Slurm, LLM, Ollama
Course Work	**PG Courses:** GPU Programming, Parallel Scientific Computing, Secure Systems Engineering, Program Analysis, Advanced Data Structures and Algorithms, Advanced Computer Architecture **UG Courses:** Operating Systems, Data Structures and Algorithms, Computer Architecture, Database Management Systems, Computer Networks, Compiler Design

Research

Capacitated Vehicle Routing Problem Jul'24 – Present

M.S. Thesis | Guide: Prof. Rupesh Nasre, Prof. Narayanaswamy N S

Context & Problem	The Capacitated Vehicle Routing Problem (CVRP) is an **NP-hard combinatorial optimization problem** which asks for a set of **minimum-cost routes** that start and end at a single depot. The routes serve a set of customers with known demands, and every vehicle has the same limited capacity. Each customer must be visited exactly once, and the total demand on a route may not exceed the vehicle capacity. CVRP sits at the core of logistics, parcel delivery and distribution planning, where even small savings in travelled distance add up to large **reductions in cost and emissions**.
Approach & Contributions	• Developed a solver built from **simple, fast heuristics**: a geometry-aware construction, granular local search, and a ruin-and-recreate metaheuristic applied through a size-independent decomposition and recombination scheme with parallel execution. • On **million-scale benchmark instances**, our solver **beats the state of the art** on average within a few seconds, and it improves best-known solutions within 600 seconds. • Achieves roughly a **10× reduction in runtime** compared to SOTA, on a standard 8-core machine.

Projects

Analyzing AI Task Deployment Workflows & Security | *Python, LangChain, Ollama, Llama 3.1* Secure Systems Engg. Coursework – Jan'26 – Apr'26

- Analyzed the enterprise risks of **excessive agency** in **Agentic AI workflows** (**OWASP Top 10**) by engineering a **LangChain** & **Llama 3.1** environment to execute a zero-click **Indirect Prompt Injection**, proving how poisoned data hijacks the **ReAct loop** for autonomous exfiltration.
- Integrated a **Human-In-The-Loop (HITL)** authorization mechanism to establish a hard security boundary, enforcing explicit user consent before high-impact tool executions to prevent unintended damage. Designed a **Dual LLM "Inspector"** architecture to function as an autonomous semantic firewall, utilizing a strict secondary model to evaluate tool parameters and neutralize malicious intent in real-time.
- Proposed an **LLM Security Gateway** and **Scoped MCP Authentication** architecture to enforce real-time semantic inspection and cryptographic integrity, mitigating multi-agent **Delegation Chain Spoofing** risks.

Domain-Specific AI Compiler & Accelerated CUDA Kernel Engine | *C++, CUDA, Triton, LLVM, CMake* May'26 – Aug'26

- Architected a domain-specific C++/Python ML compiler that parses tensor computational graphs into **MLIR/LLVM IR**, developing custom transformation passes (operator fusion, dead-code elimination, loop unrolling) to reduce graph execution overhead.
- Engineered a code generator lowering optimized IR into fused **OpenAI Triton** and **CUDA C++** kernels, applying shared memory tiling and vectorized loads (`float4`) to eliminate DRAM write-backs and runtime dispatch latency.
- Profiled hardware execution counters using **NVIDIA Nsight Compute** and **perf**, achieving a **1.42× end-to-end inference speedup** and a **38% reduction** in VRAM bandwidth consumption over PyTorch baselines.

Parallel SSSP Solver | *CUDA C++, NVIDIA Thrust* GPU Programming Coursework – Aug'25 – Nov'25

- Implemented static and adaptive **Δ-stepping parallel SSSP algorithms** on NVIDIA **GPU**s to process **ultra-large sparse graphs** in CSR format.
- Applied **Thrust** parallel algorithms to **dynamic worklist queues**, dynamically scaling bucket widths to maintain K=65,536 concurrent active work items.
- Leveraged native **atomic operations** and custom memory-reuse patterns to process 300M edges without host-device synchronization overhead.

Valley of Whispering Bamboo: Interactive 3D Portfolio | *Three.js, GLSL, Web Audio* Sep'21 – May'24

- Engineered an explorable, fully **procedural** 3D world (**21k LoC**) with **zero model, texture or audio files**, all generated in code.
- Rendered **1.7M triangles in ∼200 draw calls** via **GPU instancing** & mesh batching, with **adaptive quality** for low-end devices.
- Directed a **5-min cinematic walkthrough** mode using an autopilot and a camera director on arc-length **Bézier splines**.

Other Projects

- **PENETRALIA** – Co-engineered a real-time graphics rendering demo featuring a CUDA-FFT ocean solver, procedural terrain, atmospheric scattering, and an advanced post-processing pipeline (godrays, bloom, depth of field) guided by Bézier camera animation. **Link:** https://youtu.be/Ze5QbGQnguM
- **Asa He Chanchal Maan** – Demo created using the OpenGL graphics library and C language. Win32 SDK is used for windowing purposes and data structures in C are used to render scenes. **Link:** https://youtu.be/4pB2rZeWkNU

Positions of Responsibility & Other Activities

MS Representative	Elected representative for **MS by Research scholars** in CSE at IIT Madras; acted as liaison between students and faculty leadership on academic and administrative matters.
Teaching Assistant	Facilitated core **Systems** and **Algorithms** courses which included GPU Programming, Blockchain Technology, Natural Language Processing; designed problem sets and exams, conducted weekly tutorials, and mentored undergraduate and graduate students.
Event Coordinator	Coordinated logistics and technical operations for major IIT Madras events, including the **ICPC India Online Round** (2025, 2026) and the CSE Degree Distribution Ceremony.
NSS Head	Headed the **National Service Scheme** unit during B.Tech; directed community initiatives including blood donation drives, check-dam construction, and a 7-day rural development camp in Varoti Bk., Pune.
RSP Leader	Led the **Road Safety Patrol** platoon during high school, overseeing team drills and public traffic safety awareness drives.

Interests & Hobbies

Arts: Drawing, Painting, Calligraphy | **Music:** Tabla, Guitar | **Sports:** Table Tennis, Cricket, Volleyball, Carrom, Chess, Kho-Kho